HAND & UPPER LIMB · NEET-SS / FET / INI-SS

Nerve compression and injury

1. Nerve injury basics

Structure

- **Endoneurium** (around each axon), **perineurium** (around a fascicle: the **blood-nerve barrier** and main tensile strength), **epineurium** (around the whole nerve, plus between fascicles).
- **Myelinated fibres:** saltatory conduction at the nodes of Ranvier; **Schwann cells** (PNS).

Seddon and Sunderland classifications

Seddon	Sunderland	Damage	Tinel sign	Recovery
Neurapraxia	**I**	**Local conduction block (demyelination)**, axon intact	**Absent (no advancing Tinel)**	**Complete, within days to about 12 weeks**
Axonotmesis	**II**	**Axon disrupted, endoneurium intact**	**Advancing**	**Complete** (Wallerian degeneration, regrowth along intact tubes)
	III	Axon + **endoneurium** disrupted, perineurium intact	Advancing	**Incomplete/variable** (mismatching, scarring)

Seddon	Sunderland	Damage	Tinel sign	Recovery
	IV	Axon, endoneurium + **perineurium** disrupted; **epineurium intact** (neuroma-in-continuity)	Static (does not advance)	**None without surgery** (excise and repair/graft)
Neurotmesis	**V**	**Complete transection**	Static	**None without surgery**
	VI (Mackinnon)	**Mixed** pattern		Variable

- **Wallerian degeneration:** the distal segment degenerates (**begins within 24-48 hours**); macrophages and Schwann cells clear debris; Schwann cells form **bands of Büngner** to guide regeneration.
- **Regeneration rate:** about **1 mm/day (about 1 inch/month)** after a latent period; **advancing Tinel sign** tracks it.
- **Motor end plates degenerate irreversibly after about 12-18 months** of denervation: **motor recovery requires reinnervation before this**: hence **nerve transfers** close to the target for proximal injuries. Sensory receptors survive longer.
- **Prognosis is better with: young age**, **distal (closer to the target) injuries**, **pure motor or pure sensory nerves**, **sharp clean injuries**, **early repair**, **short gaps**; **worse with:** old age, proximal injuries, mixed nerves, crush/avulsion, delay, smoking/diabetes.

Investigations

- **Nerve conduction studies (NCS) and EMG:** wait **about 3 weeks** after injury (Wallerian degeneration must occur before denervation changes appear). **Fibrillation potentials and positive sharp waves** = denervation (axonal injury); **motor unit potentials (nascent, polyphasic)** = reinnervation (appear before clinical recovery).
- **Sensory nerve action potentials (SNAPs) preserved with numb skin = preganglionic (root avulsion) lesion** (the dorsal root ganglion cell body is

intact) (brachial plexus).
- **Ultrasound/MRI neurography:** continuity, neuroma.

Repair principles

- **Primary repair (sharp injuries)** within days; **tension-free**; **epineurial repair** (most common; align with surface vessels, fascicular pattern), **group fascicular** repair where fascicles are topographically defined (e.g. motor branch of the ulnar nerve at the wrist).
- **Gaps: nerve conduit** (hollow tube; **gaps under about 3 cm in small-calibre sensory nerves**), **processed nerve allograft** (gaps up to about 5-7 cm in sensory nerves), **autograft (sural nerve: gold standard for larger gaps)**. Avoid tension (**over about 10% elongation impairs blood flow**).
- **Nerve transfers** (e.g. **AIN to deep motor branch of ulnar nerve** (supercharge end-to-side), **Oberlin transfer** (ulnar fascicle to biceps branch), **spinal accessory to suprascapular**, **radial nerve triceps branch to axillary nerve**): convert a proximal injury into a distal one.
- **Tendon transfers** for late or irrecoverable motor loss.
- **Closed injuries with fractures (e.g. radial nerve with humeral shaft fracture):** most are neurapraxia/axonotmesis: **observe** with serial examination and EMG at about 3 months; **explore** if **no recovery by about 3-6 months**, open fractures, **nerve palsy after manipulation (relative)**, penetrating trauma.

2. Carpal tunnel syndrome (CTS)

- **Commonest compressive neuropathy.** Carpal tunnel boundaries: **floor**: carpal bones; **roof**: **transverse carpal ligament (flexor retinaculum)** from **scaphoid tubercle and trapezium** (radial) to **pisiform and hook of hamate** (ulnar). Contents: **median nerve + 9 flexor tendons (4 FDS, 4 FDP, FPL)**.
- **Associations: female**, **pregnancy**, **diabetes**, **hypothyroidism**, **rheumatoid arthritis**, **obesity**, **acromegaly**, **amyloidosis** (dialysis, **transthyretin amyloid: carpal tunnel often precedes cardiac amyloidosis**), **gout**, mass lesions, **distal radius fracture (acute CTS: urgent release if progressive)**.

- **Symptoms: nocturnal paraesthesia** in the radial three and a half digits (**palm spared**: **palmar cutaneous branch** arises about **5-6 cm proximal to the wrist crease and passes superficial to the retinaculum**), clumsiness, **thenar wasting (APB)** in severe cases.
- **Tests:**
  - **Durkan (carpal compression) test** (**most sensitive**),
  - **Phalen test** (wrist flexion 60 s),
  - Tinel sign,
  - **Semmes-Weinstein monofilament** (**most sensitive threshold test for early compression**), **two-point discrimination** (innervation density: **late** change).
  - **CTS-6** clinical diagnostic tool.
- **NCS:** prolonged **distal sensory latency** (earliest), motor latency (**over 4.5 ms**), reduced amplitudes (axonal loss); **EMG** fibrillations in APB (severe).
- **Treatment:**
  - **Night splinting in neutral**, **corticosteroid injection** (also diagnostic: **good response predicts a good surgical outcome**), treat cause.
  - **Carpal tunnel release (open or endoscopic):** similar long-term results; endoscopic: faster return to work, **higher risk of transient nerve injury**. Release **in line with the radial border of the ring finger** (avoids the **recurrent motor branch** and the **palmar cutaneous branch**; **Kaplan cardinal line**).
  - **Recurrent motor branch variants: extraligamentous recurrent (commonest, about 50%)**, **subligamentous**, **transligamentous (about 25%; at risk)**.
  - **Pillar pain** (commonest complaint after release), scar tenderness, **incomplete release** (commonest cause of persistent symptoms), recurrence, **CRPS**, nerve injury.

3. Pronator syndrome and AIN syndrome

- **Pronator syndrome:** median nerve compression in the **proximal forearm**: **sensory symptoms including the palm (palmar cutaneous**

branch territory), **no nocturnal symptoms**, **NCS usually normal**.
Compression sites (proximal to distal):

- **Ligament of Struthers** (from a **supracondylar process** of the humerus, about 5 cm above the medial epicondyle),
- **Lacertus fibrosus (bicipital aponeurosis)**,
- **Between the two heads of pronator teres** (**commonest**),
- **FDS arch (sublimis bridge)**.
- **Provocative tests:** resisted pronation with elbow extension (pronator teres), resisted elbow flexion with supination (lacertus), resisted middle finger PIPJ flexion (FDS arch).

- **Anterior interosseous nerve (AIN) syndrome: pure motor**: **FPL, FDP to index (and middle), pronator quadratus**; **cannot make an "OK" sign (pinch sign: flat pulp-to-pulp pinch)**; **no sensory loss**. Causes: compression (same sites), **Parsonage-Turner syndrome (brachial neuritis: often the true cause: severe shoulder pain preceding weakness)**, **supracondylar fracture in children (AIN is the commonest nerve injured in extension-type fractures)**. Observe for 3-6 months (most recover); explore if no recovery.

4. Cubital tunnel syndrome (ulnar nerve at the elbow)

- **Second commonest compressive neuropathy. Sites of compression (proximal to distal):**
  - **Arcade of Struthers** (about 8 cm proximal to the medial epicondyle; medial intermuscular septum),
  - **Medial intermuscular septum**,
  - **Medial epicondyle** (subluxation, valgus deformity: **tardy ulnar nerve palsy after lateral condyle non-union with cubitus valgus**),
  - **Osborne's ligament (cubital tunnel retinaculum: roof of the cubital tunnel)** (**commonest site**),
  - **Between the two heads of FCU (Osborne's fascia)**,
  - **Deep flexor-pronator aponeurosis**.
- **Elbow flexion increases pressure and stretches the nerve** (nerve elongates about 4.5-8 mm).

- **Symptoms/signs:** paraesthesia of the **little finger and ulnar half of the ring finger** + **dorsoulnar hand (dorsal cutaneous branch: distinguishes from Guyon canal compression)**, **intrinsic weakness**, **Froment sign** (FPL substitutes for **adductor pollicis**: IPJ flexion during key pinch), **Jeanne sign** (MCPJ hyperextension of the thumb), **Wartenberg sign** (**abducted little finger: unopposed EDM (radial nerve), weak third palmar interosseous**), **claw hand (ulnar claw: ring and little fingers)**, **elbow flexion test**, Tinel.
- **Ulnar paradox: proximal (elbow) lesions cause less clawing than distal (wrist) lesions** because **FDP to ring and little fingers is also paralysed** in high lesions.
- **Treatment:** activity modification, **night splint in extension (about 45 degrees)**; surgery: **in situ decompression (open or endoscopic: first-line; equivalent outcomes to transposition, fewer complications)**, **anterior transposition (subcutaneous or submuscular)** for **unstable/subluxing nerves** or after failed decompression, **medial epicondylectomy**. **Medial antebrachial cutaneous nerve** branches at risk (painful neuroma).
- **McGowan classification** (grade I sensory, II weakness, III atrophy/paralysis): severe/long-standing disease has poorer recovery.

5. Guyon canal (ulnar tunnel) syndrome

- **Boundaries: roof: volar carpal ligament (and palmaris brevis)**, **floor: transverse carpal ligament (and pisohamate/pisometacarpal ligaments)**, **radial wall: hook of hamate**, **ulnar wall: pisiform**.
- **Zones (Gross and Gelberman):**
  - **Zone 1** (proximal to the bifurcation): **motor and sensory deficit**,
  - **Zone 2** (deep motor branch, around the hook of hamate): **pure motor**,
  - **Zone 3** (superficial sensory branch): **pure sensory**.
- **Causes: ganglion (commonest)**, **hook of hamate fracture/non-union** (golfers, baseball, cyclists' palsy), **ulnar artery thrombosis/aneurysm (hypothenar hammer syndrome)**, anomalous muscles, lipoma.
- **Dorsal cutaneous branch spared** (arises **about 5-8 cm proximal to the wrist**): **dorsal hand sensation normal** (vs cubital tunnel). **FDP to**

ring/little and FCU normal.

- **Treatment:** CT/MRI to identify the cause; decompression, excision of ganglion/hamate hook.

6. Radial nerve compression

- **Radial tunnel syndrome: pain** (lateral forearm, **about 4-5 cm distal to the lateral epicondyle, over the radial tunnel**), **no motor weakness**; mimics **tennis elbow** (coexist in about 5%); **pain on resisted middle finger extension and resisted supination**; NCS normal. **Treatment:** non-operative first; decompression.
- **Posterior interosseous nerve (PIN) syndrome: motor weakness without sensory loss**: **finger and thumb extension drop**, **wrist extends in radial deviation** (ECRL (radial nerve proper) preserved; **ECU paralysed**).
- **Compression sites in the radial tunnel (FREAS):**
  - **Fibrous bands** anterior to the radiocapitellar joint,
  - **Recurrent radial vessels ("leash of Henry")**,
  - **ECRB tendinous margin**,
  - **Arcade of Frohse (proximal edge of the supinator: commonest site)**,
  - **Supinator distal edge**.
- **Other causes: Monteggia fracture-dislocation (PIN palsy: commonest nerve injury, usually neurapraxia)**, **lipoma**, **rheumatoid synovitis of the elbow** (radiocapitellar), ganglion. **Differentiate from extensor tendon rupture** (RA): **tenodesis test** (wrist flexion extends the MCPJs if tendons are intact).
- **Wartenberg syndrome (cheiralgia paraesthetica): superficial radial nerve** compression **between brachioradialis and ECRL** (tight watch, handcuffs): dorsoradial hand paraesthesia; **Tinel**, pronation with wrist ulnar deviation provokes it.
- **Saturday night palsy** (radial nerve in the **spiral groove**): **wrist drop**, **triceps spared** (branches arise proximal).

7. Other compressive and nerve lesions

- **Thoracic outlet syndrome: neurogenic (commonest, about 95%; lower trunk C8-T1: ulnar-sided symptoms, Gilliatt-Sumner hand: wasting of the thenar eminence (APB))**, **venous (Paget-Schroetter: effort thrombosis of the subclavian vein)**, **arterial (cervical rib: subclavian artery aneurysm, emboli)**. **Causes:** cervical rib, anomalous bands, scalene hypertrophy. **Tests: Roos (elevated arm stress) test**, **Adson test**, Wright. Physiotherapy first; first rib resection/scalenectomy.
- **Suprascapular nerve:**
  - **Suprascapular notch** (under the **transverse scapular ligament**): **supraspinatus + infraspinatus** weakness,
  - **Spinoglenoid notch** (**paralabral cyst** from a **SLAP tear**): **isolated infraspinatus** weakness (painless wasting: volleyball players).
- **Quadrilateral space syndrome: axillary nerve + posterior circumflex humeral artery** (fibrous bands; throwers): posterior shoulder pain, **teres minor** atrophy on MRI.
- **Long thoracic nerve palsy: medial scapular winging** (serratus anterior). **Spinal accessory nerve palsy** (posterior triangle lymph node biopsy): **lateral winging** (trapezius) with shoulder drooping.
- **Double-crush syndrome:** proximal compression (cervical radiculopathy) makes the distal nerve more susceptible to compression.

8. Tendon transfers (for nerve palsies)

- **Principles: adequate strength (lose one MRC grade after transfer: muscle of grade 5 needed)**, **expendable donor**, **similar excursion** (**wrist flexors about 33 mm, finger extensors about 50 mm, finger flexors about 70 mm**: **"3-5-7" rule**), **straight line of pull**, **one tendon one function**, **synergistic transfers easier to retrain** (wrist flexion with finger extension), **supple joints, stable soft-tissue bed**, **tissue equilibrium** (wait for swelling/scar to settle).
- **Radial nerve palsy (high): pronator teres to ECRB** (wrist extension), **FCR or FCU to EDC** (finger extension), **palmaris longus to EPL (rerouted)** (thumb extension). **Early "internal splint" (PT to ECRB)** while awaiting nerve recovery.

- **Low median nerve palsy (opposition): opponensplasty**: **EIP (Burkhalter)**, **FDS ring (Bunnell)**, **palmaris longus (Camitz: for severe CTS; mainly abduction)**, **abductor digiti minimi (Huber: congenital thumb hypoplasia)**.
- **High median:** + **ECRL to FDP index/middle**, **brachioradialis to FPL**.
- **Ulnar nerve palsy: clawing correction**: **Zancolli capsulodesis** (MCPJ volar plate), **Stiles-Bunnell (FDS lasso)**, **ECRL/ECRB transfer (Brand) to the lateral bands**; **key pinch**: **ECRB to adductor pollicis (Smith)**, **FDS ring to adductor**; **index abduction**: **EIP or APL slip to first dorsal interosseous**; **Bouvier manoeuvre** (blocking MCPJ hyperextension corrects the claw: **intact extensor mechanism**: **simple (static) procedure suffices** if positive).

Exam pearl: **Seddon/Sunderland: neurapraxia (I) no Tinel, full recovery; axonotmesis II full, III/IV incomplete/none; neurotmesis V needs surgery. Regeneration 1 mm/day. Motor end plates lost after 12-18 months: nerve transfers. EMG at 3 weeks. Preserved SNAP with anaesthetic skin = preganglionic.**

Exam pearl: **CTS: palm spared (palmar cutaneous branch); Durkan most sensitive; Semmes-Weinstein for early compression; incomplete release = commonest cause of persistent symptoms; transligamentous motor branch at risk. Pronator syndrome: palm involved, commonest between pronator teres heads. AIN: no OK sign, no sensory loss; consider Parsonage-Turner.**

Exam pearl: **Cubital tunnel: dorsal ulnar sensation lost (vs Guyon: spared); Osborne's ligament commonest; in situ decompression first-line. Ulnar paradox: distal lesions claw more. Guyon zone 2 = pure motor (hook of hamate). PIN palsy: radial deviation of the wrist on extension; arcade of Frohse; RA: tenodesis test excludes tendon rupture.**

Exam pearl: **Tendon transfers: lose one grade; 3-5-7 cm excursion; radial palsy: PT to ECRB, FCR/FCU to EDC, PL to EPL. Opponensplasty: EIP/FDS ring/Camitz (PL)/Huber (ADM).**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.